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Federated Traffic-Flow Prediction with Communication-Cost Evaluation

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Problem Description

- **Fragmented Data:** Traffic data is distributed across multiple locations.
- **Privacy Concerns:** Centralising raw traffic data creates data-sharing risks.
- **Changing Traffic Patterns:** Traffic varies across locations and time.
- **High Communication Cost:** Frequent data transfers increase network overhead.
- **Limited Collaboration:** Individual models cannot learn from patterns across locations.
- **Objective:** Predict traffic flow using Federated Learning while preserving data locality and measuring accuracy and communication cost.

Background

- Traffic prediction helps cities understand congestion and improve traffic management. However, traffic data is distributed across locations and varies over time.
- Federated Learning allows multiple traffic nodes to train models locally and share model updates with a central server. Using **Federated Averaging (FedAvg)**, the server combines these updates without collecting raw traffic data.
- **Specific Aim:** Develop a prototype that predicts traffic flow across multiple locations using Federated Learning, compares IID and Non-IID data distributions, and evaluates prediction accuracy (MAE/RMSE) and communication overhead.

Literature & RFC Review

Existing research supports Federated Learning as an approach for collaborative model training while keeping raw data decentralised.

[1] Federated Averaging (McMahan et al., 2017): Introduced FedAvg, which combines locally trained model updates. This forms the basis of our aggregation method.
[2] Advances and Open Problems in Federated Learning (Kairouz et al., 2021): Discusses challenges including heterogeneous client data, privacy, and communication efficiency.
[3] Federated Learning in Intelligent Transportation Systems (Zhang et al., 2025): Reviews applications of Federated Learning in traffic-flow prediction and identifies challenges involving data privacy, communication, and distributed traffic data.

Proposed Solution

- Distributed Data: Divide traffic data across multiple client nodes representing different locations.
- Local Training: Train a machine learning model on each node using its local traffic data.
- Privacy Preservation: Share model updates instead of transferring raw traffic data.
- Federated Aggregation: Use Federated Averaging (FedAvg) to combine local model updates at a central server.
- Global Model Updates: Distribute the aggregated global model back to client nodes.
- Performance Comparison: Compare IID and location-based Non-IID data distributions.
- Evaluation: Measure prediction accuracy using MAE and RMSE, and assess communication overhead.

Optimization Proposed

- Efficient Communication: Exchange model updates instead of transferring raw traffic data.
- Federated Averaging (FedAvg): Aggregate client model updates to build a shared global model.
- Data Distribution Comparison: Compare IID and location-based Non-IID datasets to assess the effect of data heterogeneity.
- Prediction Performance: Evaluate the model using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- Communication Overhead Measurement: Track the size of model updates exchanged during training.
- Performance Analysis: Compare centralized training (as a baseline, if implemented) with federated learning to understand accuracy and communication trade-offs.

Aspect	Before	Proposed Optimization
Data handling	Raw data may need to be centralized	Data remains at each client
Model training	Independent local models	Collaborative training through FedAvg
Data distribution	One distribution assumption	Compare IID and Non-IID scenarios
Evaluation	Prediction accuracy alone	MAE, RMSE, and communication overhead
Model improvement	Limited collaboration across locations	Global model benefits from updates across clients

Solution Architecture & Design

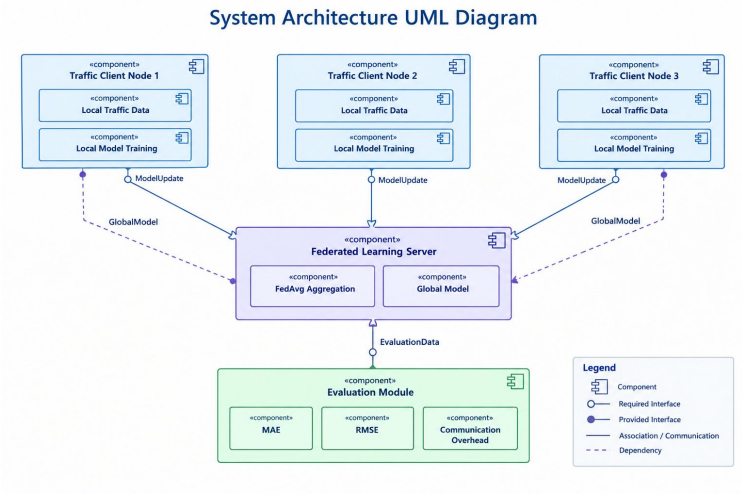


Figure: System Architecture UML Diagram

Data Flow Diagram

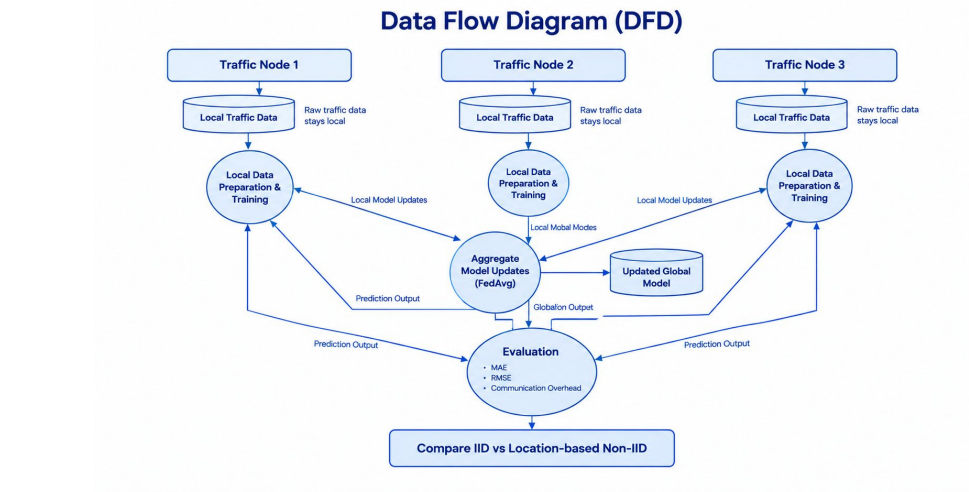


Figure: Data Flow Diagram (DFD)

Deployment Timeline

- **Phase 1** – Dataset Preparation: Inspect the Pune traffic dataset, clean the data, handle missing values, and prepare training and testing sets.
- **Phase 2** – Model Development: Develop the traffic prediction model and configure IID and location-based Non-IID client datasets.
- **Phase 3** – Federated Learning Implementation: Implement local client training and FedAvg aggregation to generate a global model.
- **Phase 4** – System Integration: Integrate the data pipeline, model, federated learning, evaluation, and communication modules.
- **Phase 5** – Testing and Evaluation: Measure MAE, RMSE, and communication overhead; compare IID and Non-IID performance.
- **Phase 6** – Finalisation: Analyse results, prepare visualisations, document the implementation, and finalise the project presentation and report.

Bibliography I

1. McMahan, B., Moore, E., Ramage, D., Hampson, S., and Agüera y Arcas, B. "Communication-Efficient Learning of Deep Networks from Decentralized Data." Proceedings of the 20th International Conference on Artificial Intelligence and Statistics, vol. 54, 2017, pp. 1273–1282. Link
2. Kairouz, P., et al. "Advances and Open Problems in Federated Learning." Foundations and Trends in Machine Learning, vol. 14, nos. 1–2, 2021, pp. 1–210. Link
3. Zhang, R., Wang, H., Li, B., and Yang, L. "A Survey on Federated Learning in Intelligent Transportation Systems." IEEE Transactions on Intelligent Vehicles, vol. 10, no. 5, 2025, pp. 3043–3059. Link

Thank You for Your Attention!